Installation

- Clone the repository from https://github.com/varunghat/circadian_soundscape
- Use a terminal with python installed and navigate to the folder (if using anaconda, use the anaconda prompt to navigate into the folder)
- Run the start_app.bat for windows, start_app.sh for linux/mac
- The app should automatically create an environment and install the requirements
- Once installed, the app should open in a web browser, while the server runs locally in the background, visible on the terminal

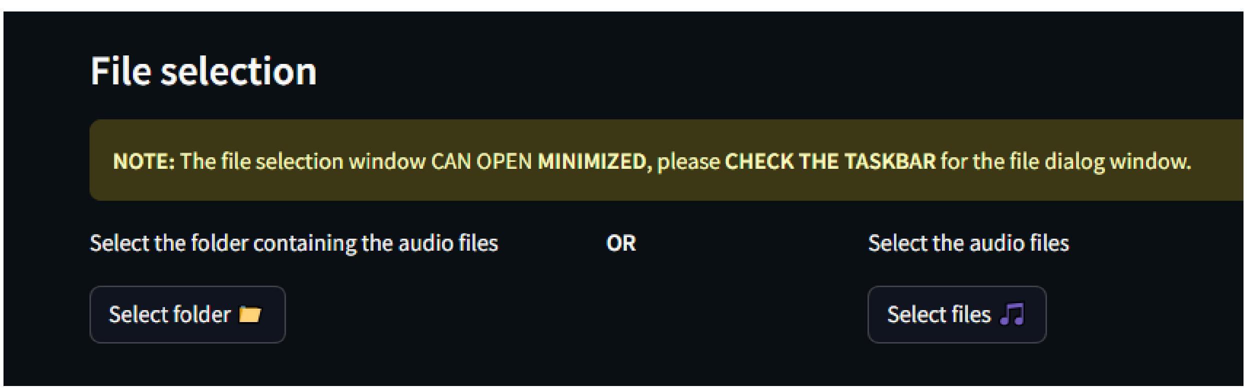
Manual installation

- Create a virtual environment (if needed) using
    - python -m venv <environment_name>
- Install requirements using
    - pip install -r requirements.txt
- Run the app using
    - streamilt run app.py

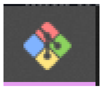
Running

File selection

The folder containing the audio files OR the audio files themselves can be selected. For now the code is tested for a day's worth of audio files. Will be improved to add multiple days, partial days, etc soon.



Once clicked, a window will open (could be minimized). Click on the minimized window and select your files OR folder.

 <- Minimized icon

Options

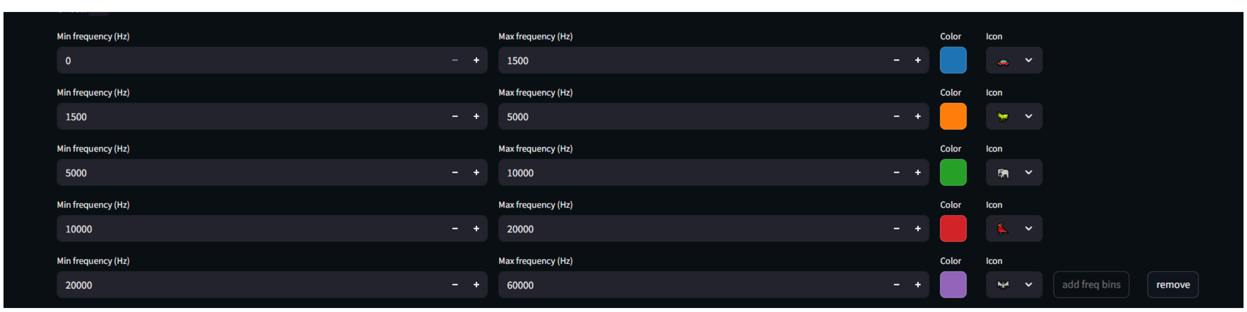
Audio format

This option allows you to select multiple audio formats based on the recording device (audiomoth, songmeter, etc) This allows the code to extract the date and time from the filenames.

Timezone offset

This option allows you to offset the time of the files if they are recorded with the wrong timezone. The offset can vary from -12 to 12, and is in hours.
NOTE: The offset is what will be **ADDED** to the time. Please enter the correction term you want.
For example:
If the recording time is behind by 3 hours from the actual time on location (eg. time was recorded in GMT but location was at GMT +3), then you need to set the offset to +3. Simple calculation is
Offset = Location time zone - recorder time zone

Frequency bins



Here you can select the frequency bins to aggregate the PMN metric over. Each bin is plotted separately, and in this section you can set the bins by setting the min Frequency and max Frequency (in Hz)
The color selected will be used to plot the particular bin. (The icons are unused for now)
To add more bins or remove bins, use the buttons **add freq bins** and **remove** respectively (visible on the last row).

Aggregation function

Here you can select the aggregation function used to aggregate the PMN results over the frequency bin. Mean is used by default.

Additional options

Sunrise sunset time

You can select this and enter the latitude and longitude of your recording location to fetch the sunrise, sunset and solar noon time at the location on the day of the recording. This will be plotted along with the PMN values to visualize patterns with respect to the daylight cycles.

Force reprocess

This option reprocesses the files even if they are already processed once. Deselect this if you have already run PMN and want to visualize the results with different aggregation, settings, etc. Enable to process the audio files again and delete previous results

Use parallel processing

This option enables parallel processing for multiple files, and you can enter the number of cores **(number of workers)** to be used. Please ensure you use less than or equal to the number of cores available on your system.

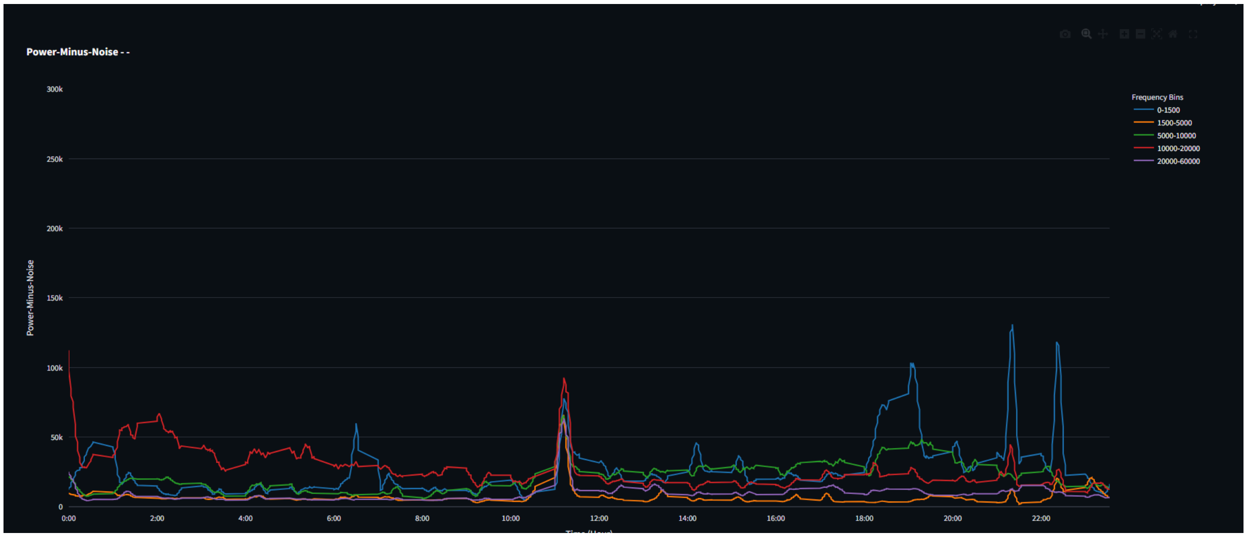
Use plotly

Select this option to plot the graphs using plotly, which allows for interactive plots. Deselect for static matplotlib plots. (Matplotlib option will ideally be removed in the future for simplicity)

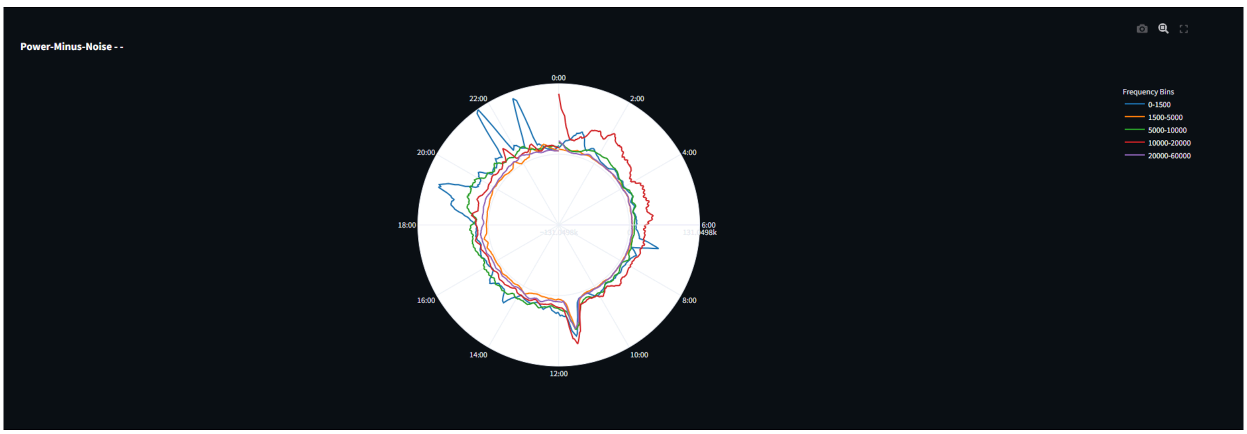
Visualizing

Press the **Visualize** button to start processing the files and visualize the plots. Note, calculating PMN is computationally intensive and will take time based on the audio file size, sampling rate and length.
Aggregation and visualization should happen instantaneously.

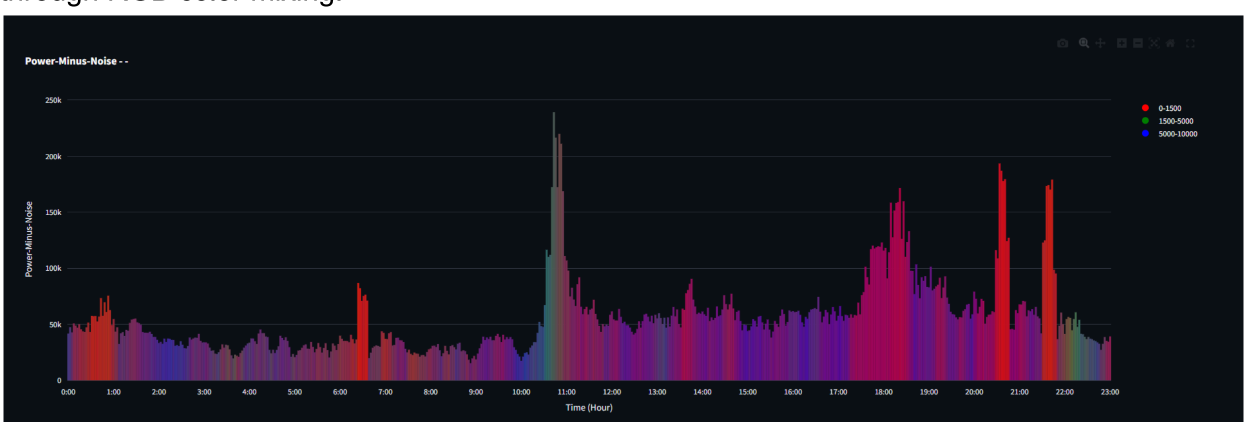
The graphs are in the following order:
1) Aggregated power-minus-noise line plot. You can select each bin to visualize separately by using the legend on the right. You can zoom in and out using the options on the top-right of the plot. You can also reset the scale and download the plots from the same options.



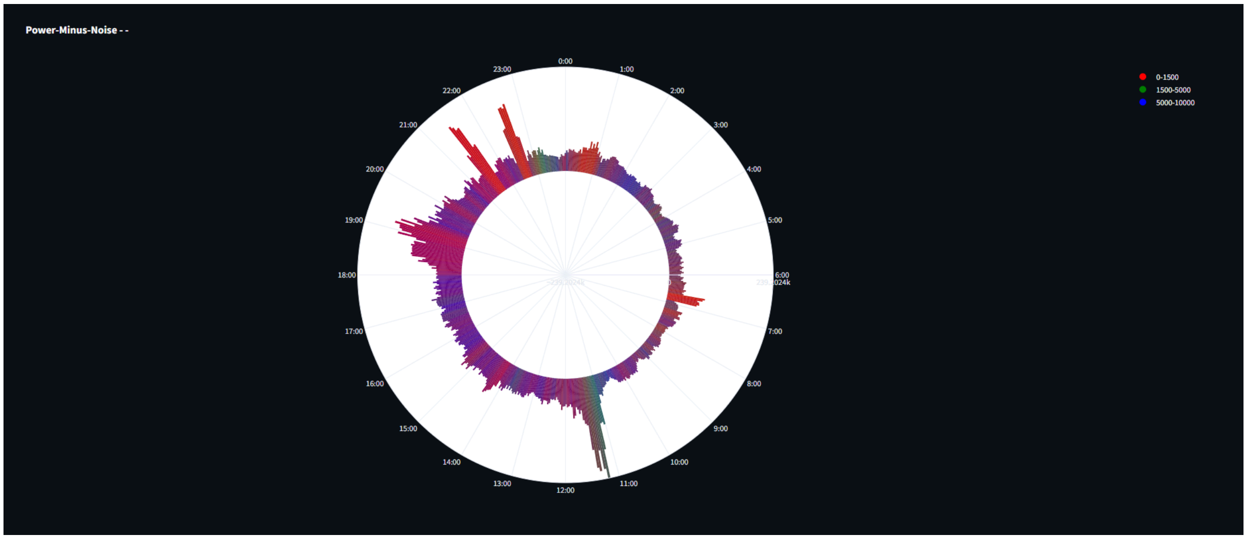
2) Polar plot of aggregated PMN. This is the same plot as the above, but displayed in a polar format which shows the cyclical nature of the data.



3) Aggregated PMN colored bar plot. Here you can visualize which bins overlap in which areas through RGB color mixing.



4) Aggregated PMN colored polar bar plot



Closing the server

The server will keep running even if the tab on the browser is closed. Please ensure to end the server by closing the terminal OR using the close server option in the webpage. By closing the webpage, you can open it again as the app will be running in the background. Background running without any process will not use much memory or processing.

Issues and Feature requests

Please report any issues, or feature requests by creating an issue on github here:
https://github.com/varunghat/circadian_soundscape/issues

You can also email the team to report the issue, if greater detail and correspondence is needed.